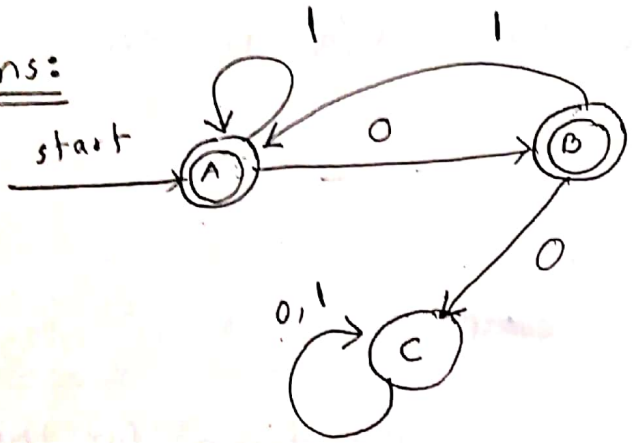
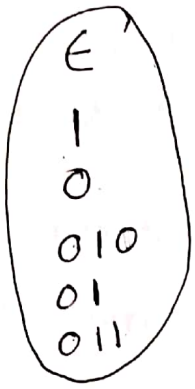
	0	1
→ *A	B	A
*B	C	A
C	C	C

Describe the language accepted by this DFA.

Ans:





This DFA supports all the strings, which ~~have~~ may have at least one zero at any position.

Non-Deterministic Finite Automata (NFA)

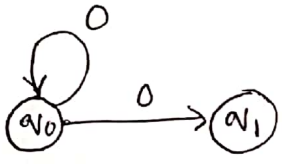
A Non-deterministic finite automation (NFA) has the power to be in several states at once.

Definition of NFA:

$$NFA = (Q, \Sigma, \delta, q_0, F)$$

1. Finite set of states, Q
2. Input symbols Σ
3. Transition function δ

$$\delta(q_0, 0) = \{q_0, q_1\}$$

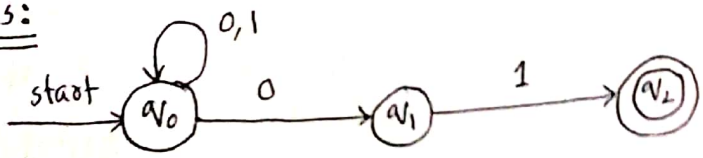


[may return more than one state]

4. starting state q_0
5. A set of final or accepting states F.

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Ans:



Here, $Q = \{q_0, q_1, q_2\}$

$\Sigma = \{0, 1\}$

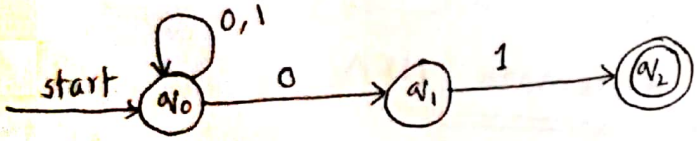
$\delta(q_0, 0) = \{q_0, q_1\}$, $\delta(q_0, 1) = \{q_0\}$

$\delta(q_1, 1) = q_2$

starting state q_0

$F_\bullet$ = Final state $\{q_2\}$

Ques: show the string 1101 is valid string for the following NFA.



Ans:

$\delta(q_0, 1) = \{q_0\}$

$\delta(q_0, 11) = \delta(\delta(q_0, 1), 1)$

$= \delta(q_0, 1)$

$= \{q_0\}$

$\delta(q_0, 110) = \delta(\delta(q_0, 11), 0)$

$= \delta(\{q_0\}, 0)$

$= \{q_0, q_1\}$

$\delta(q_0, 1101) = \delta(\delta(q_0, 110), 1)$

$= \delta(\{q_0, q_1\}, 1)$

$= \delta(q_0, 1) \cup \delta(q_1, 1) = \{q_0\} \cup \{q_2\}$

$= \{q_0, q_2\}$

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